

#10.18(a)

| Percentage of Parents with College |        |
|------------------------------------|--------|
| =====                              |        |
| Parent Percentage_with_College     |        |
| -----                              |        |
| 1 Mother                           | 12.150 |
| 2 Father                           | 11.682 |
| -----                              |        |

#10.18(b)

| Correlation Matrix |       |            |            |
|--------------------|-------|------------|------------|
| =====              |       |            |            |
|                    | educ  | MOTHERCOLL | FATHERCOLL |
| -----              |       |            |            |
| educ               | 1     | 0.359      | 0.398      |
| MOTHERCOLL         | 0.359 | 1          | 0.355      |
| FATHERCOLL         | 0.398 | 0.355      | 1          |
| -----              |       |            |            |

The correlations show positive relationships between the parent's college status and the wife's education. A logical argument for using the dummy variables (MOTHERCOLL, FATHERCOLL) over continuous years of education is that obtaining some college experience represents a structural socioeconomic shift or milestone, which might be a more distinct predictor of a child's educational attainment than simply adding one more year of high school.

#10.18(c)

| 95% CI for EDUC   |        |                   |
|-------------------|--------|-------------------|
| =====             |        |                   |
|                   | Model  | Lower_95 Upper_95 |
| -----             |        |                   |
| 1 IV (MOTHERCOLL) | -0.001 | 0.153             |
| -----             |        |                   |

#10.18(d)

| First-Stage F-test Result |             |
|---------------------------|-------------|
| =====                     |             |
| Test                      | F_statistic |
| -----                     |             |
| 1 F-test for MOTHERCOLL   | 63.563      |
| -----                     |             |

The F-statistic for the hypothesis that MOTHERCOLL has no effect on EDUC is **63.563**. This value is greater than 10, so we consider MOTHERCOLL to be a **strong instrument**.

#10.18(e)

| 95% CI Comparison |          |          |  |
|-------------------|----------|----------|--|
| =====             |          |          |  |
| Model             | Lower_95 | Upper_95 |  |
| -----             |          |          |  |
| 1 IV (Mother)     | -0.001   | 0.153    |  |
| 2 IV (Both)       | 0.028    | 0.148    |  |
| -----             |          |          |  |

Using both instruments, the 95% interval estimate is typically **narrower** than the one in part (c) because adding another valid instrument increases the efficiency of the estimation.

#10.18(f)

| Joint First-Stage F-test  |             |
|---------------------------|-------------|
| =====                     |             |
| Test                      | F_statistic |
| -----                     |             |
| 1 Joint F-test (Both IVs) | 56.963      |
| -----                     |             |

The joint F-statistic for MOTHERCOLL and FATHERCOLL is **59.963**. This value is safely above the rule-of-thumb threshold of 10 which means that these instruments are considered **adequately strong**.

#10.18(g)

| Surplus Instrument Validity   |         |
|-------------------------------|---------|
| Test                          | p_value |
| 1 Sargan (Overidentification) | 0.626   |

The Sargan overidentification test yields a p-value of **0.626**.  $\therefore p > 0.05$ , we fail to reject the null hypothesis and conclude that the surplus instrument is **valid**.

#10.20(a)

| OLS CAPM                          |                         |
|-----------------------------------|-------------------------|
| Dependent variable:               |                         |
| y                                 |                         |
| x                                 | 1.202***<br>(0.122)     |
| Constant                          | 0.003<br>(0.006)        |
| Observations                      | 180                     |
| R2                                | 0.352                   |
| Adjusted R2                       | 0.349                   |
| Residual Std. Error               | 0.081 (df = 178)        |
| F Statistic                       | 96.804*** (df = 1; 178) |
| Note: *p<0.1; **p<0.05; ***p<0.01 |                         |

The estimated beta for Microsoft is **1.202**.  $\therefore \text{beta} > 1$ , Microsoft stock is classified as **risky** relative to the market portfolio.

#10.20(b)

| First-Stage Results (RANK) |      |       |   |
|----------------------------|------|-------|---|
| Variable R_squared p_value |      |       |   |
| 1                          | RANK | 0.913 | 0 |

**IV1 & IV3:** It clearly satisfies IV1 (the rank itself does not directly affect Microsoft's return) and IV3 (it is highly correlated with the market return by construction).

**IV2:** The exogeneity condition (IV2) is debatable because RANK is derived from a variable with measurement error. However, using ranks is a classic method to smooth out measurement errors because ranks are less sensitive to specific measurement deviations than the raw data itself, making it potentially valid.

The coefficient of RANK is very significant ( $p < 0.001$ ). The  $R^2$  of the first-stage regression is **0.913**. Because the relationship between the actual market return and its rank is mechanically strong, RANK can be regarded as a **strong IV** in terms of relevance.

#10.20(c)

| Augmented Regression Test                       |         |
|-------------------------------------------------|---------|
| Test                                            | p_value |
| 1 Significance of $v_{\text{hat}}$ (Exogeneity) | 0.043   |

The coefficient of the first-stage residual  $v_{\text{hat}}$  has a p-value of **0.043**.  $\because$  p-value  $> 0.01$ , we **do not reject** the null hypothesis of exogeneity at the 1% level, concluding that the market return is exogenous.

#10.20(d)

| OLS vs IV (RANK)                  |                         |                     |
|-----------------------------------|-------------------------|---------------------|
| Dependent variable:               |                         |                     |
|                                   | y                       |                     |
|                                   | OLS                     | instrumental        |
|                                   | OLS                     | variable            |
|                                   | (1)                     | IV (RANK)           |
|                                   | (1)                     | (2)                 |
| x                                 | 1.202***<br>(0.122)     | 1.278***<br>(0.128) |
| Constant                          | 0.003<br>(0.006)        | 0.003<br>(0.006)    |
| Observations                      | 180                     | 180                 |
| R2                                | 0.352                   | 0.351               |
| Adjusted R2                       | 0.349                   | 0.347               |
| Residual Std. Error (df = 178)    | 0.081                   | 0.081               |
| F Statistic                       | 96.804*** (df = 1; 178) |                     |
| Note: *p<0.1; **p<0.05; ***p<0.01 |                         |                     |

The IV estimate of beta is **1.278**. Measurement error in the explanatory variable typically causes "attenuation bias," pushing the OLS estimate toward zero. As expected, the IV estimate is **larger in magnitude** than the OLS estimate, correcting this downward bias.

#10.20(e)

| First-Stage Results (RANK & POS) |             |           |
|----------------------------------|-------------|-----------|
| Test                             | F_statistic | R_squared |
| 1 Joint F-test (RANK, POS)       | 951.262     | 0.915     |

The joint F-statistic for RANK and POS is **951.262**, and the  $R^2$  is **0.915**. Since the F-statistic is well above the conventional threshold of 10, we conclude that we have **adequately strong IVs**.

#10.20(f)

| Augmented Regression Test (Both IVs) |         |
|--------------------------------------|---------|
| =====                                |         |
| Test                                 | p_value |
| -----                                |         |
| 1 Significance of v_hat_e            | 0.029   |
| -----                                |         |

Testing the new residual  $v\_hat\_e$ , the p-value is **0.029**.  $\therefore p > 0.01$ , we do not reject the null hypothesis and conclude that the market return is **exogenous** at the 1% level of significance.

#10.20(g)

| OLS vs IV (Both)               |                             |                     |
|--------------------------------|-----------------------------|---------------------|
| =====                          |                             |                     |
|                                | Dependent variable:         |                     |
|                                | -----                       |                     |
|                                | y                           |                     |
|                                | OLS                         | instrumental        |
|                                | OLS                         | variable            |
|                                | (1)                         | IV (Both)           |
|                                | (2)                         |                     |
| -----                          |                             |                     |
| x                              | 1.202***<br>(0.122)         | 1.283***<br>(0.128) |
| Constant                       | 0.003<br>(0.006)            | 0.003<br>(0.006)    |
| -----                          |                             |                     |
| Observations                   | 180                         | 180                 |
| R2                             | 0.352                       | 0.351               |
| Adjusted R2                    | 0.349                       | 0.347               |
| Residual Std. Error (df = 178) | 0.081                       | 0.081               |
| F Statistic                    | 96.804*** (df = 1; 178)     |                     |
| =====                          |                             |                     |
| Note:                          | *p<0.1; **p<0.05; ***p<0.01 |                     |

The IV estimate using both instruments is **1.283**. Similar to part (d), this estimate corrects for the measurement error attenuation bias and is larger than the OLS estimate, which **agrees with our expectations**.

#10.20(h)

| Surplus Instrument Validity |           |         |
|-----------------------------|-----------|---------|
| =====                       |           |         |
| Test                        | Statistic | p_value |
| -----                       |           |         |
| 1 Manual Sargan Test        | 0.558     | 0.455   |
| -----                       |           |         |

The manual Sargan test yields a statistic of **0.558** with a p-value of **0.455**. If the p-value is > 0.05, we fail to reject the null hypothesis, concluding that the surplus instrument is **valid** .